

**Theorem 0.1 (LU decomposition):** Let  $A_{n \times n}$  be a square matrix. If all  $k$  leading principal minors of  $A$  are non-singular (for  $k = 1, \dots, n-1$ ), then  $A$  has a unique factorization

$$A = LU,$$

where  $L_{n \times n}$  and  $U_{n \times n}$  are lower and upper triangular matrices respectively.

**Explicit formula:** The elements of  $L$  and  $U$  are computed as follows:

$$\begin{aligned} L_{ii} &= 1, \quad \text{for } i = 1, \dots, n \\ U_{ij} &= A_{ij} - \sum_{k=1}^{i-1} L_{ik} U_{kj}, \quad \text{for } j \geq i \\ L_{ij} &= \frac{1}{U_{jj}} \left( A_{ij} - \sum_{k=1}^{j-1} L_{ik} U_{kj} \right), \quad \text{for } i > j \end{aligned}$$